

Also see Kozeny-Carmen

$$\frac{\Delta P}{L} = - \frac{180\mu}{\Phi^2 D_p^2} \frac{(1-\epsilon)^2}{\epsilon^3} v_s$$

[SAME CONCLUSIONS AS ASSESSMENT BELOW]

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(sphericity of bed particles ($\Phi=1$ for spheres))

Bed level estimation.

The Ergen equation expresses friction factor of a packed column.

$$f_p = \frac{150}{Gr_p} + 1.75$$

Packing equivalent diameter (as if a sphere).

ΔP of bed

$$f_p = \frac{\Delta P}{L} \frac{D_p}{\rho v_s^2} \left(\frac{\epsilon^3}{1-\epsilon} \right)$$

Voidage fraction.

$$Gr_p = \frac{\rho v_s D_p}{(1-\mu)\epsilon} = \frac{Re}{(1-\epsilon)}$$

Packed bed friction factor

length of bed (not column)

Liquid density.

superficial fluid velocity.

fluid viscosity.

$$\text{so } L = f(\Delta P, D_p, \rho, v_s, \epsilon, \mu)$$

Calculable.

low accuracy as both temperature and concn dependent.

Measured BUT depends on temp and concn and bed height.

Measured

⚠ Very difficult to predict. Undermines this estimate.

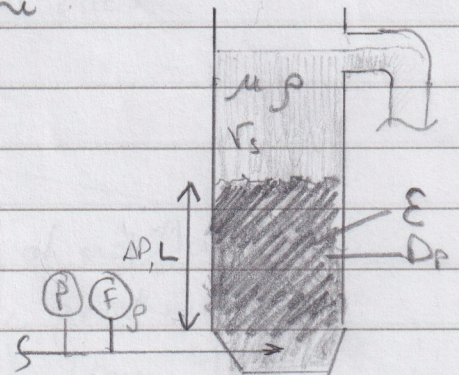
Dependent on input material.

⚠ Not easy to predict. Changes with input material which is not consistent.

Calculable from measured flowrate.

This is dynamic and $\therefore$ inaccurate.

Because this needs to subtract ^{static} pressure of liquid above the solid, which CHANGES.



This method will give a very poor estimate of solid level. Not considered viable.